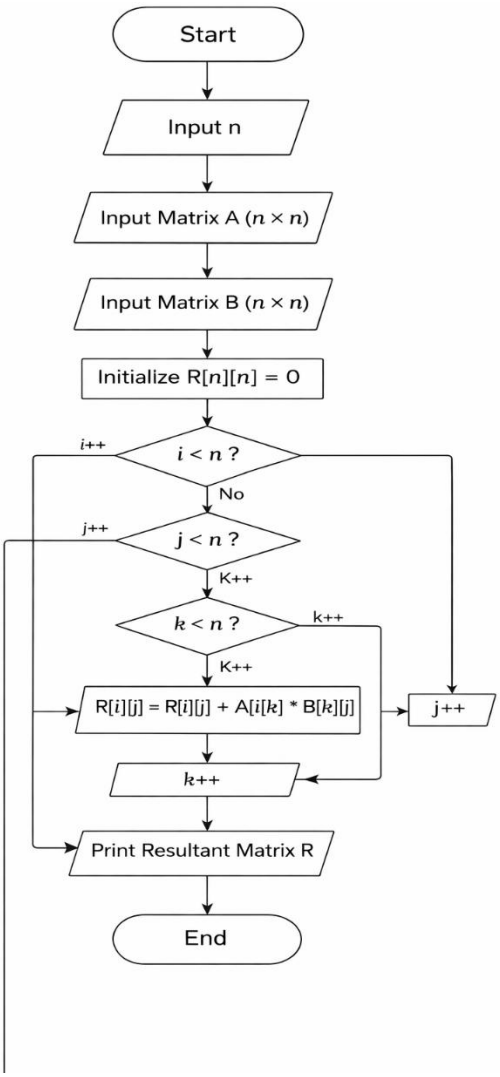
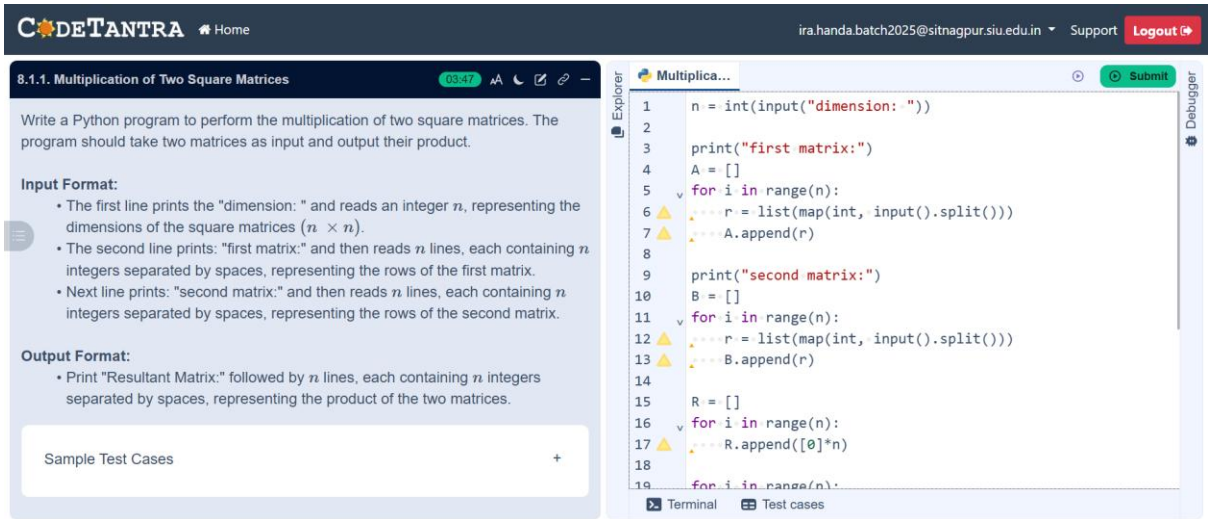
EXPERIMENT 8.1.1

FLOWCHART:



CODETANTRA:



ALGORITHM:

1. **Start**
2. **Input n**
   Read the dimension of the matrices (n × n).
3. **Input first matrix A**

Read n rows

Each row contains n integers

4. **Input second matrix B**

Read n rows

Each row contains n integers

5. **Initialize result matrix R**

Create an empty matrix R of size n × n

Set all elements of R to 0

6. **Multiply matrices**

or each row i from 0 to n-1:

For each column j from 0 to n-1:

Set R[i][j] = 0

For each k from 0 to n-1:

Multiply A[i][k] * B[k][j]

Add it to R[i][j]

7. **Display result**

Print "Resultant Matrix:"

Print matrix R row by row

8. **End**

EXPERIMENT 8.2.1

CODETANTRA:

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8.1.2. Prime Numbers in a Given Range 03:51

Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).

Input Format:

- First line contains an integer representing the start of the range.
- Second line contains an integer representing the end of the range.

Output Format:

- Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a separate line.
- If no prime numbers exist in the range, print: "No primes".

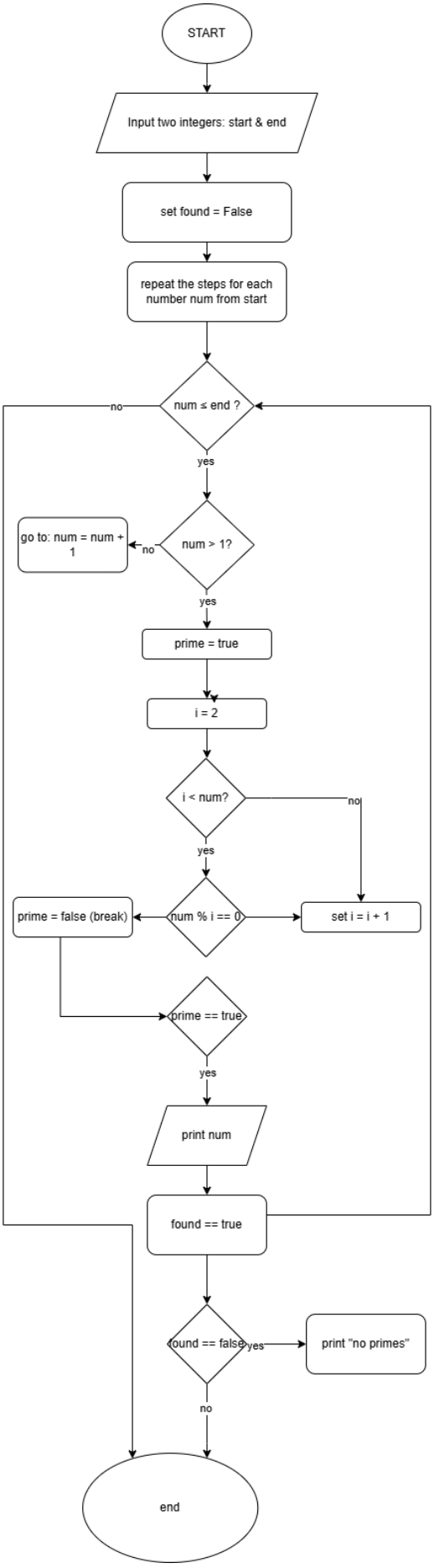
Sample Test Cases +

Explorer primeNu... Submit Debugger

```python
start = int(input())
end = int(input())
found = False
for num in range(start, end + 1):
    if num > 1:
        prime = True
        for i in range(2, num):
            if num % i == 0:
                prime = False
                break
        if prime:
            print(num)
            found = True

if not found:
    print("No primes")
```

Terminal Test cases


START
Input two integers: start & end
set found = False
repeat the steps for each number num from start
num ≤ end ?
no
yes
num > 1?
no
go to: num = num + 1
yes
prime = true
i = 2
i < num?
no
yes
num % i == 0
prime = false (break)
set i = i + 1
prime == true
yes
print num
found == true
found == false
yes
print "no primes"
no
end

FLOWCHART:

ALGORITHM:

1. Start

2. Input two integers start and end.

3. Initialize a variable found and set it to False.

Repeat the following steps for each number num from start to end:

a. Check if num is greater than 1.

b. If yes, then:

4. Set prime = True.

ii. Repeat for each integer i from 2 to num - 1:

If num is divisible by i (i.e., num % i == 0), then:

5.Set prime = False

Exit the loop

6. If prime == True, then:

- Print the value of num
- Set found = True

7. After completing the loop, check:

- If found == False, then print "No primes"

 End